10 Complex FDE Scenarios

Foundation & Design Engineering — Case Study Workshop

01 DPDP × GenAI Data Principal Rights Agent for Real Estate

A real-estate company wants an AI agent to handle consent withdrawal, erasure requests, correction requests, access requests, WhatsApp opt-outs, grievance routing, and nominee-related requests. The agent must check CRM, lead portals, broker-uploaded files, campaign platforms, call-center notes, consent logs, and retention rules before preparing a response.

Participant Question: *Is the main problem prompt, context, harness, or loop?*

Best Answer

Primary:	Secondary:
Harness Engineering	Context Engineering + Loop Engineering

Why: This is not just answering a privacy question. The agent must verify identity, inspect consent state, check retention obligations, trigger downstream workflows, preserve evidence, route grievances, and escalate exceptions. The DPDP Act gives Data Principals rights including access, correction, completion, updating, and erasure, and the 2025 Rules operationalize rights handling and response expectations.

⚠ **FDE Trap:** Calling this a "DPDP chatbot" is too shallow. It is a rights-handling workflow agent.

02 Banking AI Credit-Memo Copilot

A bank wants an AI copilot that drafts credit memos using borrower financials, bureau reports, GST data, relationship history, covenant status, collateral valuation, repayment behaviour, sector outlook, and early-warning signals. The output must separate facts, assumptions, model-derived risk indicators, policy exceptions, and relationship-manager judgment.

Participant Question: *Which layer fails if the memo sounds fluent but misses covenant breaches?*

Best Answer

Primary:	Secondary:
(to be filled)	(to be filled)

Why: (to be filled)

⚠ **FDE Trap:** (to be filled)

03 SOC Phishing Triage Agent with Controlled Tool Access

A SOC team wants an AI agent that inspects suspicious emails, checks headers, detonates URLs in a sandbox, queries threat intelligence, correlates with SIEM alerts, checks identity-risk signals, assigns severity, and recommends containment.

Participant Question: *Which engineering layer is most critical before production?*

Best Answer

Primary:	Secondary:
(to be filled)	(to be filled)

Why: (to be filled)

⚠ **FDE Trap:** (to be filled)

04 Manufacturing Quality Deviation Investigator

A factory wants AI to investigate recurring product defects. The agent must read MES logs, sensor drift, batch records, operator notes, maintenance history, supplier lots, calibration records, control-chart deviations, and previous CAPA actions. It should identify likely root causes and recommend investigation paths.

Participant Question: *Is this just RAG over quality documents?*

Best Answer

Primary:	Secondary:
(to be filled)	(to be filled)

Why: (to be filled)

⚠ **FDE Trap:** (to be filled)

05 AI Software Migration Agent for Legacy Modernization

An enterprise wants to migrate a legacy Java/Spring application to a newer cloud-native architecture. The agent must inspect code, understand architecture patterns, update dependencies, modify tests, run builds, fix errors, check security findings, prepare a pull request, and generate migration evidence.

Participant Question: *Which layer decides whether this becomes useful or chaotic?*

Best Answer

Primary:	Secondary:
(to be filled)	(to be filled)

Why: (to be filled)

⚠ **FDE Trap:** (to be filled)

06 Insurance Claims Fraud Investigation Copilot

An insurer wants AI to detect suspicious claims by comparing claim narratives, hospital invoices, policy exclusions, previous claims, provider history, image metadata, payment patterns, and external fraud indicators. It should recommend "pay", "investigate", or "refer to human adjudicator", but not autonomously reject claims.

Participant Question: *Which layer protects the organization from unfair or unsupported decisions?*

Best Answer

Primary:	Secondary:
(to be filled)	(to be filled)

Why: (to be filled)

⚠ **FDE Trap:** (to be filled)

07 Enterprise Procurement Negotiation Copilot

A procurement team wants AI to analyze vendor proposals, benchmark pricing, detect risky clauses, compare SLAs, flag data-processing obligations, check security terms, and draft negotiation positions. It must use approved templates and route high-risk clauses to legal, privacy, security, or finance.

Participant Question: *Where does prompt engineering help, and where does it fail?*

Best Answer

Primary:	**Secondary:**
(to be filled)	(to be filled)

Why: (to be filled)

⚠ **FDE Trap:** (to be filled)

08 Healthcare Clinical Documentation Assistant

A hospital wants AI to draft discharge summaries from doctor notes, lab reports, medication lists, procedures, imaging reports, allergies, and follow-up instructions. It must flag contradictions, missing instructions, and possible medication conflicts for clinician review.

Participant Question: *Which layer is responsible for detecting missing or contradictory clinical facts?*

Best Answer

Primary:	**Secondary:**
(to be filled)	(to be filled)

Why: (to be filled)

⚠ **FDE Trap:** (to be filled)

09 AI Governance Evidence Collector for ISO 42001 / EU AI Act Readiness

A company wants an AI governance assistant that maps AI use cases to risk classes, collects evidence, links model cards, captures DPIA/AIA documents, tracks approvals, records evaluation results, manages supplier attestations, and flags missing controls before audit.

Participant Question: *Is this mainly prompt engineering or something else?*

Best Answer

Primary:	**Secondary:**
(to be filled)	(to be filled)

Why: (to be filled)

⚠ **FDE Trap:** (to be filled)

10 Telco Network Incident Agent

A telco wants an AI agent that investigates network incidents by reading alarms, topology, ticket history, change records, device logs, customer-impact maps, SLA commitments, and recent configuration changes. It should identify likely fault domains and recommend rollback, dispatch, failover, or escalation.

Participant Question: *Which layer matters most when alarms are noisy and incomplete?*

Best Answer

Primary:	Secondary:
(to be filled)	(to be filled)

Why: (to be filled)

⚠ **FDE Trap:** (to be filled)